

Regex for Log Parsing and Analysis Report

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Indicators of Compromise (IoCs) and Analysis Goals

Purpose of Analysis:

The purpose of this analysis is to parse raw server logs to identify potential security incidents, specifically focusing on unauthorized access attempts and suspicious user behavior. By extracting specific patterns, we aim to convert unstructured log data into actionable intelligence.

Why detecting these IoCs is critical:

- **Failed Logins:** Repeated failures often indicate a Brute Force or Credential Stuffing attack where an adversary is trying to guess passwords.
- **Unauthorized Access (403/401):** Attempts to access restricted areas (like `/admin` or `/sensitive_data`) suggest an attacker is scanning the web server for vulnerabilities or unsecured endpoints.
- **Odd-Hour Activity:** High-privilege actions (like root access) occurring between 5 PM and 9 AM are often indicative of malicious activity or compromised insider accounts, as they fall outside standard monitoring windows.

Key IoCs Investigated:

1. **Source IP Addresses:** To identify the origin of attacks.
 2. **Failed Login Messages:** To detect authentication attacks.
 3. **Unauthorized Access Warnings:** To detect privilege escalation attempts.
 4. **Timestamps (Odd Hours):** To establish a timeline of suspicious activity.
 5. **Sensitive Record Access:** To track potential data exfiltration.
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Regex Patterns and Data Extraction

Regex Patterns Constructed

IoC Type	Regex Pattern Used	Explanation
IP Addresse s	<code>\b\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}\b</code>	<code>\b</code> ensures word boundaries. <code>\d{1,3}</code> matches 1 to 3 digits (0-999). <code>\.</code> matches the literal dot separator. Repeats 4 times for IPv4 format.
Failed Logins	<code>`.*(Failed login</code>	Login attempt failed
Unauthor ized Access	<code>`.*(Unauthorized access</code>	403).*`
Odd Hours	<code>`[\d{4}-\d{2}-\d{2}]s(0[0-8]</code>	1[7-9]
Sensitive Records	<code>`.*(Access to record ID</code>	Record accessed: ID)\s(\d+)`

Data Extracted

Pattern	Extracted Data (Samples from Log)
Suspicious IPs	192.168.1.100, 10.0.0.1, 10.10.10.10, 203.0.113.50
Failed Logins	Failed login for user eve, Failed login for user bob, Login attempt failed for admin, Failed login for root, Login failure for admin
Unauthorized Access	Unauthorized access attempt by user root, Unauthorized access by user_unknown, Unauthorized access attempt admin, Unauthorized access user unknown
Timestamps (Odd)	02:45:12, 23:10:15, 18:05:05, 03:15:45, 04:10:00, 06:22:30, 20:45:10, 02:50:20, 00:30:45, 23:55:00
Record IDs	98765 (User alice), 12345 (User charlie), 67890 (User alice)

Analysis of Results

Summary of Findings:

- Targeted Root Attack:** The IP 192.168.1.100 is aggressively targeting the root account between 02:45 AM and 03:15 AM. The logs show unauthorized access attempts to /admin followed by failed logins.
- Persistent Admin Attack:** A different IP, 10.10.10.10, is conducting a brute force attack against the admin account during evening hours (20:45 PM and 15:10 PM).
- Insider Threat / Compromised Host:** The IP 10.0.0.1 (likely an internal network address) is probing sensitive endpoints (/sensitive_data, /admin) and failing logins for user eve at irregular hours (06:22 AM, 18:05 PM).
- Suspicious Data Access:** User alice (IP 203.0.113.50) is accessing record IDs 98765 and 67890. While the status is 200 (Success), the access occurs late at night (23:10 PM and 19:30 PM), which warrants verification.

Insights and Recommendations:

The data suggests a coordinated attack from external IPs (192..., 10.10...) attempting to gain administrative control. Simultaneously, there is potentially malicious behavior originating from inside the network (10.0.0.1).

- **Immediate Action:** Block IPs 192.168.1.100 and 10.10.10.10 at the firewall.
 - **Priority Investigation:** Isolate the machine 10.0.0.1 and audit user eve's activity.
 - **Policy Check:** Verify if user alice has a business justification for accessing records at 11 PM.
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Reflection on Regex Process

Effectiveness of Regex:

Using regex transformed a manual reading task into a precise data extraction process. Instead of reading every line, regex allowed me to instantly filter out "noise" (like valid traffic from charlie or jane) and focus purely on errors and threats. It standardized the output, making it easier to spot the correlation between timestamps and failed logins.

Challenges and Adjustments:

- **Log Inconsistency:** The biggest challenge was the variation in log messages. For example, some logs said "Failed login for..." while others said "Login failure for...".
- **Adjustment:** I had to refine my regex from a simple string match to using a Group/OR operator (Option A|Option B) to ensure I didn't miss the variations.
- **Time Ranges:** Creating a regex for "odd hours" was complex because simply matching numbers \d would catch valid hours (like 14:00). I had to create specific ranges (e.g., 0[0-8]) to accurately target non-business hours.

Broader Implications:

Mastering regex is essential for modern cybersecurity because attackers generate massive amounts of data. Manual analysis is impossible at scale. Regex serves as the foundation for automated rules in SIEMs (like Splunk or Wazuh). By defining these patterns now, we can create automated alerts that trigger instantly when these specific regex conditions are met, shifting our security posture from reactive (analyzing old logs) to proactive (real-time defense).